

NTSS Booth Floor Plan System

Project Report

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Project: NTSS Booth Floor Plan System

Language: Java

Interface: Java Swing

1. Introduction

The NTSS Booth Floor Plan System is a desktop application designed to help NTSS staff place and manage booths on a floor plan. The application provides a graphical interface where staff can select a booth shape and size, place the booth on the floor plan, remove booths, and save the floor plan.

2. System Design

The application is divided into model, controller, and user-interface responsibilities. FloorPlan represents the floor plan and owns the collection of booths. Booth represents one booth. BoothGroup stores booth components as a composite structure. FloorPlanController handles operations requested by the interface. NTSSStaffUI and FloorPlanCanvas provide the graphical interface.

3. Design Patterns

Controller: FloorPlanController handles operations between the interface and model. Creator: FloorPlan creates Booth objects. Expert: FloorPlan manages information about its booths. Composite: BoothComponent, Booth, and BoothGroup provide a common component structure. Iterator: BoothIterator and FloorPlanIterator provide sequential booth access. Flyweight: BoothType and BoothTypeFactory share booth type data for matching shape and size combinations.

4. User Interface

The Swing interface provides shape and size selections, palette buttons, a floor plan canvas, status information, and save/remove controls. The canvas responds to mouse clicks and checks booth placement before adding a booth.

5. Booth Placement

A booth is checked against the floor plan boundaries and existing booths before placement. Shape affects drawing and size affects booth dimensions.

6. Saving

The Save option uses a file chooser and writes the current floor plan information to a text file, including the floor plan dimensions and placed booths.

7. Testing

The implementation was checked for application startup, booth selection, valid placement, boundary rejection, overlap rejection, booth removal, saving, repeated booth types, iteration, and booth groups.

8. Files

The project includes the Java source files, README, .gitignore, and this report.

9. Conclusion

The completed system provides booth floor plan functionality while separating responsibilities between the model, controller, iterator, composite, flyweight, and Swing interface classes.

Testing Summary

Test	Expected Result
Start application	Main window opens
Place booth	Booth is added and drawn
Boundary check	Invalid booth is rejected
Overlap check	Overlapping booth is rejected
Remove booth	Selected booth is removed
Save	Floor plan is written to a file
Iterator	Placed booths can be traversed